

Title

Effects of humor-based aerobic activity on psychological distress and physical function in lung cancer patients undergoing chemotherapy: protocol for a single-blinded parallel-group trial

Abstract

Background: Patients with advanced lung cancer experience substantial psychological burden and physical deconditioning during chemotherapy. Novel, accessible interventions combining exercise with positive psychological engagement may reduce distress and improve functional outcomes. **Methods:** This single-blind, parallel-group randomized trial will randomize 84 lung cancer patients undergoing active chemotherapy to an intervention group receiving structured humor-based movement sessions or a control group receiving standard oncology care. The intervention group will participate in specialized sessions combining rhythmic movement and laughter-inducing activities twice weekly. Primary outcomes include measured psychological distress using validated stress scales and psychological resource assessment. Secondary outcomes include exercise capacity testing and walking tolerance. Measurement points are baseline and immediately following the intervention period. Block randomization will be employed with concealment of allocation from assessors.